

Paras Grammar School

Class 4

Chapter 1: Exploring Windows

Answer the following questions.

a. What is an operating system?

Operating system is a computer program which manages all resources. Windows is one of the most popular operating systems in the world.

b. Why Operating system is crucial for a computer's functioning?

An operating system is the most essential part of your computer. Windows is one of the most popular operating systems in the world. Without it, your computer could not function. You wouldn't be able to run programs or control the hardware.

c. Describe an instance where you used Windows 10 on a tablet. How did the touch-friendly features enhance your experience?

I used Windows 10 for drawing shapes. Using Windows 10 on a tablet is a fun and easy experience because of its touch-friendly features.

d. Recall a situation where you had to use the Task Manager to resolve an issue. What steps did you take, and what was outcome?

I used Task Manager to restart the hangman game. I followed following steps to resolve the issue:

Step 1: Open *Task Manager*

Step 2: Look for the software you want to restart

Step 3: Right click the application

Step 4: Click *Restart*

- e. Consider a time when you switched between Tablet mode and Desktop model on a Windows 10 device. How did this feature benefit you?**

I switched between Tablet and Desktop mode when I want to use touch gestures. This features saved my cost because I don't have to buy a separate touchscreen device.

- f. What steps can you take to troubleshoot common problems in Windows 10? Explain why these steps are effective.**

Troubleshoot resolved a lot of issues in my computer. I don't have to repair my computer from shop instead I use troubleshoot option by following these steps:

Step 1: Open Settings application in Windows

Step 2: Go to Troubleshoot Settings

Step 3: Look for available measures

Chapter 2: Email Communication

Answer the following questions.

a. What is the purpose of a subject line in an email?

The *subject line* in an email is a short sentence that tells the recipient what the email is about. It serves as the title of email.

b. Why it is essential to have a clear subject line?

A clear subject line in an email is essential because it helps the recipient understand the purpose of the email at a glance.

c. Describe the process of attaching files to an email in Gmail.

We can attach files to an email by following the steps:

Step 1: After completing your message, click on the paperclip icon at the bottom of the window

Step 2: The file upload dialog will appear. Locate and select the file you want to attach from your device or cloud storage.

Step 3: Wait for the attachment to upload.

Step 4: Click *Send* after attachment is uploaded.

d. What is the importance of organizing your emails using labels in Gmail?

Using labels in Gmail helps in organizing relevant mails under one label. We can move important emails as reminder under our label. Label helps in arranging our mails which saves a lot of time from searching for important mails.

e. Explain what BCC (Blank Carbon Copy) is in an email and when you might use it.

BCC option in email helps in hiding the other recipients which helps in managing security and professionalism. I will use this option when I don't want all recipient know about each other about the received mail.

Chapter 3: Creating Digital Design with Canva

Answer the following questions.

a. What are the tools available in the object panel in Canva?

Object panel contains the following tools:

1. Design
2. Elements
3. Text
4. Uploads
5. Draw

b. What is negative space, and why is it important in design?

Negative space is the **empty space** around and between design elements, such as text, images, and shapes. It does not have to be white—it can be any color. It makes text and content easier to read. A well balanced design looks more attractive and modern.

c. How can you collaborate on a design project using Canva?

With Canva, you can collaborate on design projects with your colleagues. Share your design work via links or email to work together on the same task. We can share design by following steps:

Step 1: Open the design file and click the *Share* option at the top of the editor

Step 2: Type the email address of the participant

Step 3: Select permission type: *Edit* or *View*

d. How would you create a birthday e-card in Canva?

I will create a birthday e-card by following the steps:

Step 1: Open Canva

Step 2: On the left-hand side, choose the template

Step 3: Change the text written

Step 4: Change the background color

e. How you can effectively use negative space to improve the visual appeal of your design?

To effectively use negative space, follow these guidelines:

1. Keep it simple and uncluttered
2. Use proper spacing in text
3. Create focus and emphasis
4. Balance the design
5. Choose background wisely

Chapter 4: Fundamentals of Design Thinking

Answer the following questions.

a. What is the importance of empathy in the Design Thinking process?

Empathy is important in design thinking process because we are designing for someone else whose needs, wants and behavior matters. Understanding the user helps in getting valuable insights which enhances design decision.

b. Why is iteration essential in the design thinking process?

Iteration is essential in design thinking process because it delivers the design in small increments which focus on small part of entire design.

c. Describe the central theme in a mind map.

The central theme is the main title of the mind map. For instance, if you are looking to organize a birthday party, the central theme would be 'Birthday Party'.

d. Describe the associations in a mind map.

Associations emerge with ideas that connect to the central theme. In the case of a birthday party, the associations can be birth date, location, gifts, activities, etc.

e. Apply the Design thinking process to a personal challenge.

Consider the time management problem in studies for design thinking approach as follows:

1. **Empathize:** understand why managing study time is difficult by identifying distractions and challenges
2. **Define:** Clearly state the problem, like struggling with focus due to social media.

3. **Ideate:** Brainstorm different ways to improve study habits, such as using a planner or timers.
4. **Prototype:** Try one solution, like following a study schedule with breaks.
5. **Test:** Use the plan, check if it works, and improve it if needed.

f. How does the Design Thinking process help in solving complex problems?

Design thinking process helps in solving complex problems by dividing the problem into different parts. Each part focuses on a single part of the problem. By connecting these solved parts, entire problem is solved.

Chapter 5: Cloud Computing

Answer the following questions.

a. What is Google Drive?

Google Drive is a well-known cloud-based service that allows you to create, store, and organize data. It also offers free online programs to create documents, spreadsheets, and other types of files.

b. How Google Drive can be used to organize and manage files?

Google Drive can be used to organize and manage files by using following features:

1. Create Folders
2. Search Easily
3. Backup Important Files
4. Share and Collaborate
5. Access Anywhere

c. Describe the process of joining the Zoom meeting.

You can join a Zoom meeting by clicking on the meeting link provided through email or message. When you click the link, Zoom will ask if you want to attend the meeting from a web browser or Zoom app. You can choose any option and join the meeting.

d. How can you protect yourself from common Internet threats?

We can protect our self from Internet threats by following the guidelines:

1. Keep you information confidential
2. Beware of what you share on the internet
3. Check your settings
4. Never agree to meet strangers in real life

e. What are the benefits of using Google Drive for file organization and management?

Following are some benefit of using Google Drive for file organization and management:

1. Ease of Use
2. Recovery of data
3. Ease of Access
4. Ease of Sharing

Chapter 6: Introduction to Coding and AI

Answer the following questions.

a. Name three common programming languages.

Following are some of the popular programming languages:

1. Python
2. Java
3. Ruby

b. How does machine learning help computers learn from data?

Machine learning helps computers learn from data. Following are the approaches used:

1. Collect Data
2. Identify Patterns
3. Make Predictions
4. Improve Over Time

c. How do AI systems provide users with recommendations?

AI system analyze user data and performs following steps for recommendations:

1. Collect Data
2. Find Patterns
3. Make Predictions
4. Show Recommendations

d. Name the features that make Google Maps interactive.

Following features make Google Maps interactive:

1. Search Bar
2. Zoom In and Out
3. Street View

4. Directions and Navigations

5. Live Traffic Updates

e. Write an example using Event block in Sprite Lab.

Consider making a character jump when space is pressed:

1. Create a cat sprite

2. Use the “When key pressed” event block

3. Choose “space” as the key

4. Add a “jump” action

f. Create a flowchart to illustrate the process of training a machine learning model.

